Forecasting Volatility with Spillover Effects: Perspectives from Graph Neural Networks

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Outline

Introduction

Preliminaries

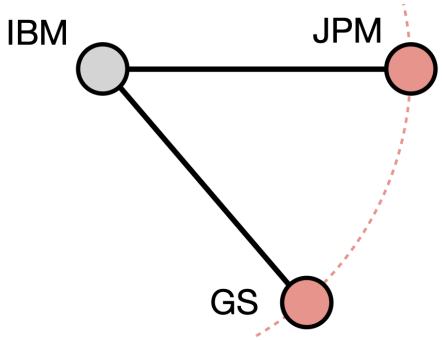
Proposed Methodology

Empirical Analysis

Conclusion

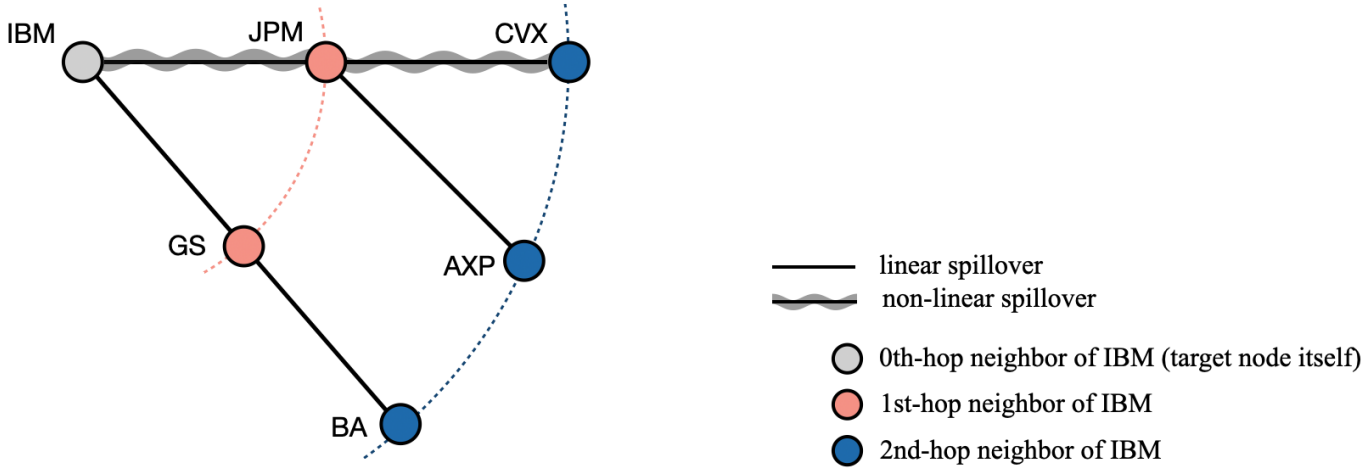
Volatility Modeling and Forecasting

- Risk Management, Derivative Pricing, Trading Strategies ...
- Realized Volatility (RV): sum of squared intraday returns [Andersen et al., 2003]
  - Heterogeneous Autoregressive (HAR) [Corsi, 2009]
    $RV_t = \alpha + \beta_d RV_{t-1} + \beta_w RV_{t-5:t-2} + \beta_m RV_{t-22:t-6} + u_t$
- Volatility Spillover Effects:



  - It can benefit the understanding and forecasting of volatilities ([Buncic and Gisler, 2016, Degiannakis and Filis, 2017]).

New Lens through Graphs and ML



Questions

- 1. *Multi-hop neighbors*
- 2. *Nonlinear relations*
- 3. *Estimation criterion (EC)*

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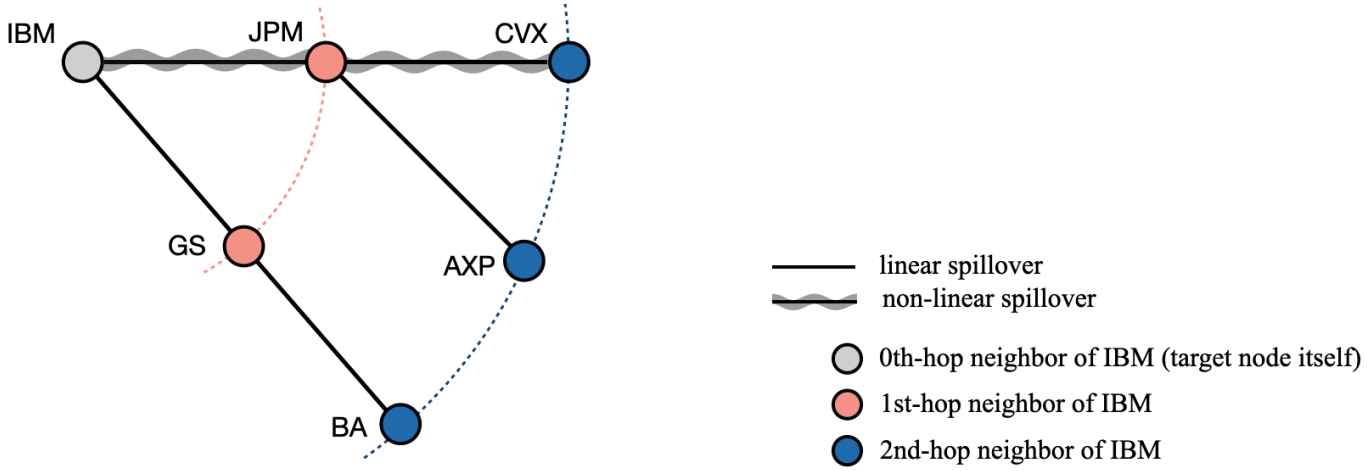

Questions

- 1. *Multi-hop neighbors*
- 2. *Nonlinear relations*
- 3. *Estimation criterion (EC)*

Machine learning: $y = F_\theta(\boldsymbol{x})$

- predictor set $\boldsymbol{x}$
- model specification F
- loss function y v.s. $F_\theta(\boldsymbol{x})$, optimization

Main Contributions

- We put forward a methodology to forecast multivariate RVs by incorporating spillover effects via **graph neural networks (GNNs)**.

Takeaway:

1. Multi-hop information is not necessary, as long as 0th-hop and 1st-hop are included.
2. Nonlinear GNN model significantly improves the forecasting performance of linear counterparts.
3. Choosing Quasi-likelihood (QL) as the training loss function can significantly improve the predictive accuracy, compared to the usual MSE.
   - QL-trained models $\leftrightarrow$ Multiplicative Error Model (MEM) $\leftrightarrow$ error heteroskedasticity $\leftrightarrow$ weights to observations.

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Graph Definitions

Definition (Graphs)

A graph $\mathcal{G}$ is defined as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V} = \{v_1, \ldots, v_N\}$ is a set of N nodes and $\mathcal{E}$ is a set of edges, where $e_{ij} = (v_i, v_j) \in \mathcal{E}$ denotes an edge connecting node v_i and node v_j.

Definition (Adjacency matrix)

An adjacency matrix $\mathbf{A}$ is a square matrix whose dimension is $N \times N$, where $\mathbf{A}[i, j]$ represents the connection between v_i and v_j in the graph $\mathcal{G}$. If $\mathbf{A}[i, j] \in \{0, 1\}$, $\forall i, j$, the graph is a binary graph.[1] In this paper, we focus on **binary** graphs.

Example (Two Trivial Graphs)

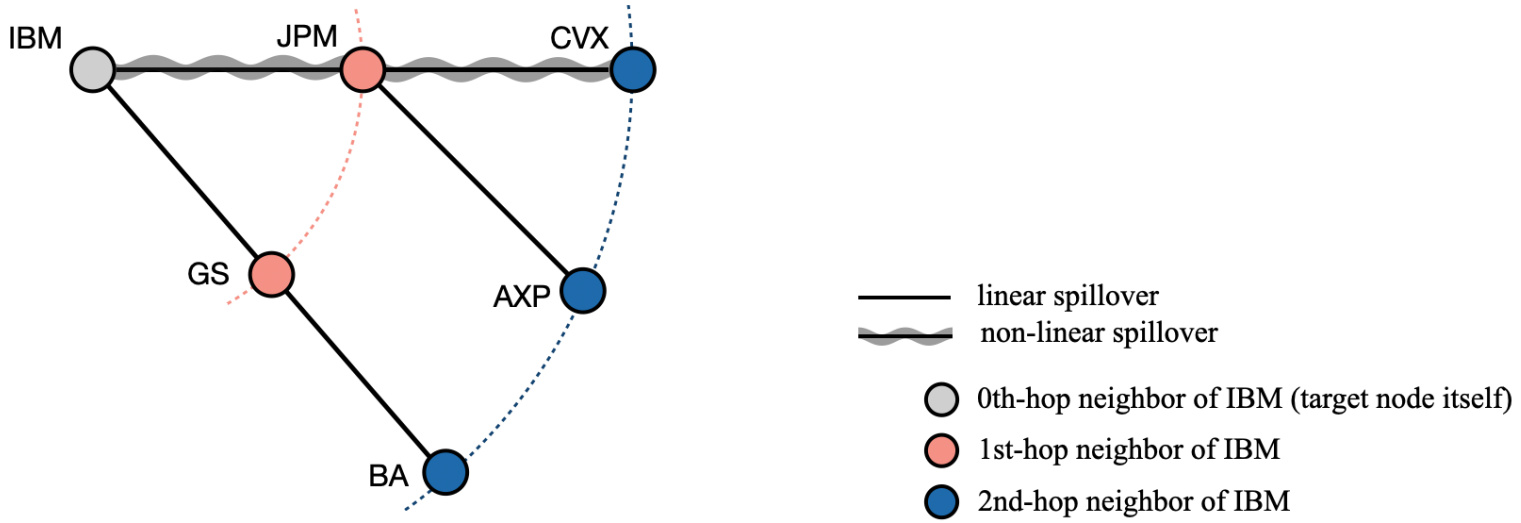
Complete graph: all ones except along the diagonal where there are only zeros.

Empty graph: a zero matrix.

[1] An adjacency matrix can be weighted, where $\mathbf{A}[i, j] \geq 0$, $\forall i, j$ represents the strength/intensity of the connection between nodes v_i and v_j.

Graph Definitions (cont.)

Definition (kth-hop neighbors)



Following [Feng et al., 2022], we use the ***k*th-hop neighbors** to represent the nodes with exact distance k from node v, based on the shortest path distance (SPD) kernel.

Graph Construction

Graphical LASSO (GLASSO) is a sparsity-penalized maximum likelihood estimator for the precision matrix ($\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$ i.e. inverse of covariance matrix) proposed by [Friedman et al., 2008]. GLASSO is to obtain $\hat{\boldsymbol{\Theta}}$ such that:

$$\hat{\boldsymbol{\Theta}} = \text{argmin}_{\boldsymbol{\Theta} \succeq 0} \left(\text{Tr}(\boldsymbol{S}\boldsymbol{\Theta}) - \log\det(\boldsymbol{\Theta}) + \lambda \sum_{j \neq k} |\boldsymbol{\Theta}_{jk}| \right)$$

where $\boldsymbol{S}$ is the sample covariance of asset returns, and λ is the penalizing parameter to control the sparsity of $\hat{\boldsymbol{\Theta}}$.

- If the ij-th component of the precision matrix is zero, then i-th and j-th variables are conditionally independent, given the other variables.
- The adjacency matrix $\boldsymbol{A}$ from GLASSO is defined as $\boldsymbol{A}[i,j] = 1$ if $\hat{\boldsymbol{\Theta}}[i,j] \neq 0$; otherwise $\boldsymbol{A}[i,j] = 0$.

Graph HAR

Denote $\boldsymbol{v}_t = (RV_{1,t}, \ldots, RV_{N,t})'$ as the vector of cross-sectional realized volatilities, **neighborhood aggregation** on graphs ([Zhang et al., 2022]), denoted as **GHAR**, is:

$$\begin{aligned}\mathbb{E}(\boldsymbol{v}_t|\mathcal{F}_{t-1}) &= \boldsymbol{\alpha} + \beta_d \boldsymbol{v}_{t-1} + \beta_w \boldsymbol{v}_{t-5:t-2} + \beta_m \boldsymbol{v}_{t-22:t-6} \\ &\quad + \gamma_d \boldsymbol{W} \cdot \boldsymbol{v}_{t-1} + \gamma_w \boldsymbol{W} \cdot \boldsymbol{v}_{t-5:t-2} + \gamma_m \boldsymbol{W} \cdot \boldsymbol{v}_{t-22:t-6}, \\ &= \boldsymbol{\alpha} + \underbrace{\boldsymbol{V}_{:t-1}\boldsymbol{\beta}}_{Self} + \underbrace{\boldsymbol{W}\boldsymbol{V}_{:t-1}\boldsymbol{\gamma}}_{Graph},\end{aligned} \tag{1}$$

- $\boldsymbol{\alpha} \in \mathbb{R}^N, \boldsymbol{\beta} = (\beta_d, \beta_w, \beta_m)' \in \mathbb{R}^3, \boldsymbol{\gamma} = (\gamma_d, \gamma_w, \gamma_m)' \in \mathbb{R}^3$ are parameters to be estimated.
- $\boldsymbol{V}_{:t-1} = [\boldsymbol{v}_{t-1}, \boldsymbol{v}_{t-5:t-2}, \boldsymbol{v}_{t-22:t-6}] \in \mathbb{R}^{N\times 3}$.
- $\boldsymbol{W} = \boldsymbol{O}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{O}^{-\frac{1}{2}}$ is the normalized adjacency matrix without self-connections.
- $\boldsymbol{O} = \mathrm{diag}\{n_1, \ldots, n_N\}$, where $n_i = \sum_j \boldsymbol{A}[i,j], \forall i$.

Graph Convolutional Network (GCN)

GCN ([Kipf and Welling, 2017]) approximates the graph convolution with the first-order Chebyshev spectral polynomials.

$$\boldsymbol{H}^{(l+1)} = \sigma\left(\tilde{\boldsymbol{O}}^{-\frac{1}{2}}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{O}}^{-\frac{1}{2}}\boldsymbol{H}^{(l)}\boldsymbol{\Theta}^{(l)}\right), \tag{2}$$

where $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}_N$ is the adjacency matrix of the graph $\mathcal{G}$ with added self-connections.

- $\tilde{\boldsymbol{O}}$ is a diagonal matrix with $\tilde{\boldsymbol{O}}_{ii} = \sum_j \tilde{\boldsymbol{A}}_{ij}$.
- $\tilde{\boldsymbol{O}}^{-\frac{1}{2}}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{O}}^{-\frac{1}{2}}$ is the normalized adjacency matrix to stabilize the backpropagation in training the models.
- $\boldsymbol{H}^{(l)} \in \mathbb{R}^{N\times D^{(l)}}$ is the matrix of node representations at l-th layer. $\boldsymbol{H}^{(0)}$ is the input node features, e.g. $\boldsymbol{V}_{:t-1}$.
- $\boldsymbol{\Theta}^{(l)} \in \mathbb{R}^{D^{(l)}\times D^{(l+1)}}$ is the layer-specific trainable weight matrix.
- $\sigma(\cdot)$ denotes a nonlinear activation function, such as $\mathrm{ReLU}(\cdot) = \max(0, \cdot)$.
- The mechanism of an individual stock's past volatility on its future volatility is mixed with the spillover effect.

CNN vs GCN

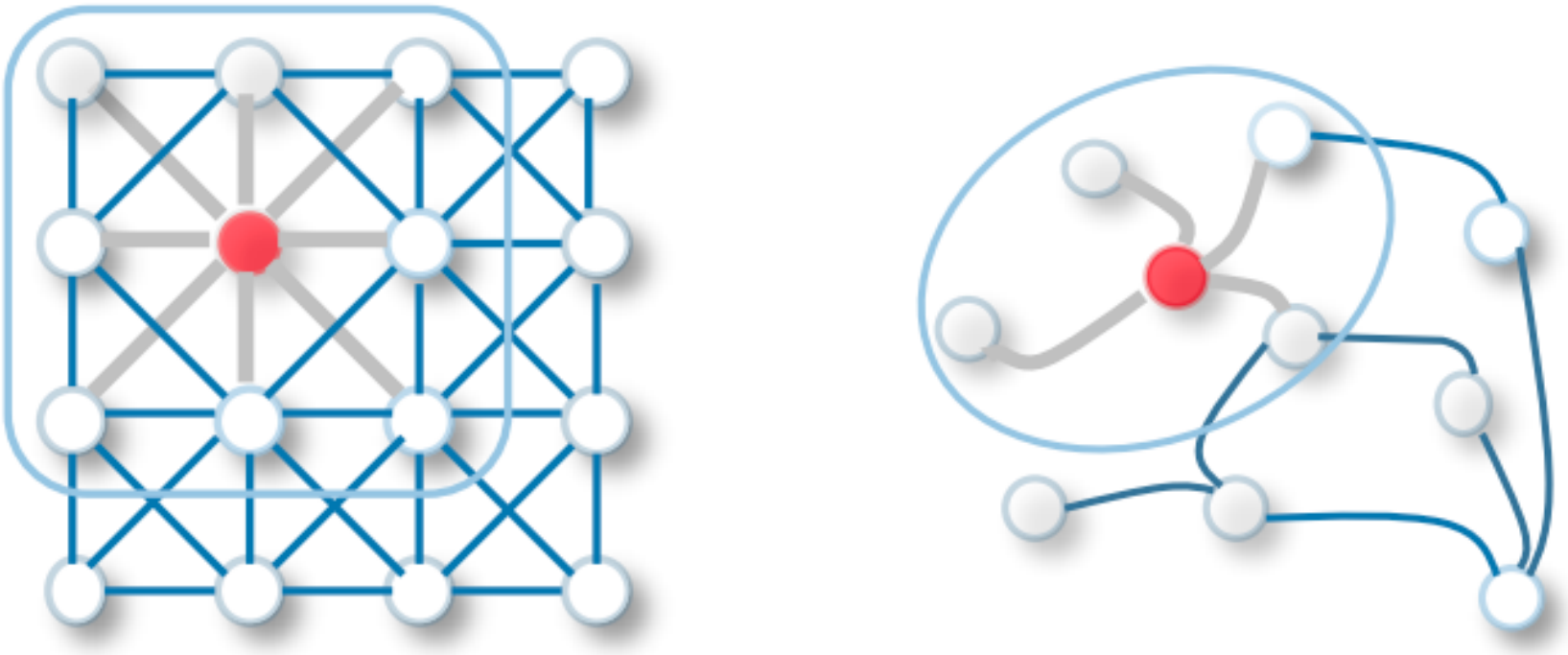


Figure 1: Convolution Neural Network (CNN) vs GCN

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GNN-enhanced HAR (GNNHAR)

GNN propagation rule:

$$\mathbf{GNN}(\boldsymbol{H}^{(l)}, \boldsymbol{A}): \quad \boldsymbol{H}^{(l+1)} = \mathrm{ReLU}\left(\boldsymbol{O}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{O}^{-\frac{1}{2}}\boldsymbol{H}^{(l)}\boldsymbol{\Theta}^{(l)}\right). \quad (3)$$

GNN-enhanced HAR model:

$$\mathbf{GNNHAR1L}(\boldsymbol{A}): \quad \begin{aligned} \boldsymbol{H}^{(1)} &= \mathrm{GNN}(\boldsymbol{V}_{:t-1},\ \boldsymbol{A}) \\ \mathbb{E}(\boldsymbol{v}_t|\mathcal{F}_{t-1}) &= \boldsymbol{\alpha} + \boldsymbol{V}_{:t-1}\boldsymbol{\beta} + \boldsymbol{H}^{(1)}\boldsymbol{\gamma} \end{aligned} \quad (4)$$

$$\mathbf{GNNHAR2L}(\boldsymbol{A}): \quad \begin{aligned} \boldsymbol{H}^{(2)} &= \mathrm{GNN}(\boldsymbol{H}^{(1)},\ \boldsymbol{A}) \\ \mathbb{E}(\boldsymbol{v}_t|\mathcal{F}_{t-1}) &= \boldsymbol{\alpha} + \boldsymbol{V}_{:t-1}\boldsymbol{\beta} + \boldsymbol{H}^{(2)}\boldsymbol{\gamma} \end{aligned} \quad (5)$$

$$\mathbf{GNNHAR3L}(\boldsymbol{A}): \quad \begin{aligned} \boldsymbol{H}^{(3)} &= \mathrm{GNN}(\boldsymbol{H}^{(2)},\ \boldsymbol{A}) \\ \mathbb{E}(\boldsymbol{v}_t|\mathcal{F}_{t-1}) &= \boldsymbol{\alpha} + \boldsymbol{V}_{:t-1}\boldsymbol{\beta} + \boldsymbol{H}^{(3)}\boldsymbol{\gamma} \end{aligned} \quad (6)$$

GHAR with nonlinearity (GNNHAR)

We first develop a custom GNN propagation rule:

$$\mathbf{GNN}(\boldsymbol{H}^{(l)}, \boldsymbol{A}): \quad \boldsymbol{H}^{(l+1)} = \mathrm{ReLU}\left(\boldsymbol{O}^{-\frac{1}{2}}\boldsymbol{A}\boldsymbol{O}^{-\frac{1}{2}}\boldsymbol{H}^{(l)}\boldsymbol{\Theta}^{(l)}\right). \tag{7}$$

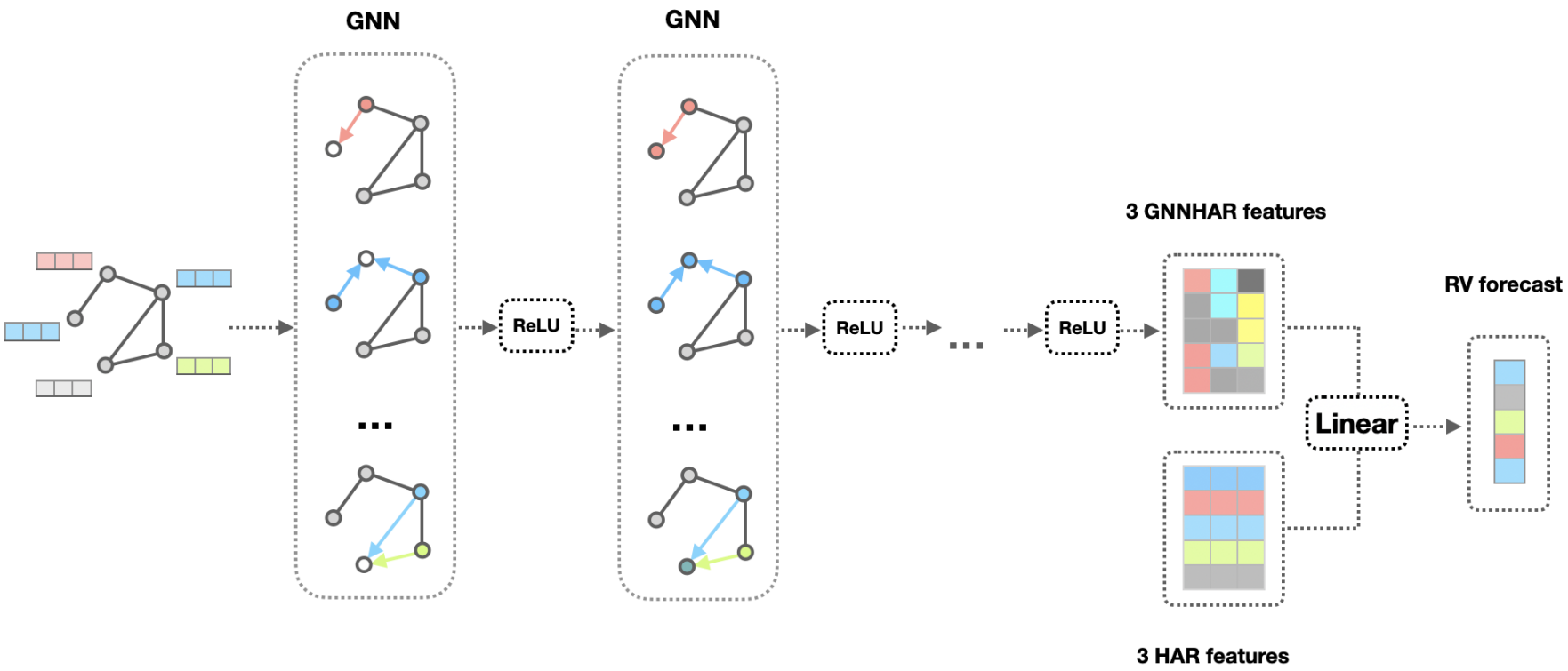
Replace the linear neighborhood aggregation in GHAR with the proposed GNN layer, we develop a GNN-enhanced HAR model.

$$\begin{aligned} \boldsymbol{H}^{(1)} &= \mathrm{GNN}(\boldsymbol{V}_{:t-1},\ \boldsymbol{A}) \\ \mathbf{GNNHAR1L}(\boldsymbol{A}): \quad \boldsymbol{v}_t &= \boldsymbol{\alpha} + \boldsymbol{V}_{:t-1}\boldsymbol{\beta} + \boldsymbol{H}^{(1)}\boldsymbol{\gamma} + \boldsymbol{u}_t. \end{aligned} \tag{8}$$

$$\begin{aligned} \boldsymbol{H}^{(2)} &= \mathrm{GNN}(\boldsymbol{H}^{(1)},\ \boldsymbol{A}) \\ \mathbf{GNNHAR2L}(\boldsymbol{A}): \quad \boldsymbol{v}_t &= \boldsymbol{\alpha} + \boldsymbol{V}_{:t-1}\boldsymbol{\beta} + \boldsymbol{H}^{(2)}\boldsymbol{\gamma} + \boldsymbol{u}_t. \end{aligned} \tag{9}$$

$$\begin{aligned} \boldsymbol{H}^{(3)} &= \mathrm{GNN}(\boldsymbol{H}^{(2)},\ \boldsymbol{A}) \\ \mathbf{GNNHAR3L}(\boldsymbol{A}): \quad \boldsymbol{v}_t &= \boldsymbol{\alpha} + \boldsymbol{V}_{:t-1}\boldsymbol{\beta} + \boldsymbol{H}^{(3)}\boldsymbol{\gamma} + \boldsymbol{u}_t. \end{aligned} \tag{10}$$

Architecture



[Multi-hop effects] After K layers of graph convolution in a GNN model, every node representation is determined by the information from the nodes within K hops (see [Feng et al., 2022]).

Estimation Criterion (EC)

- Mean squared error (MSE): $\left(RV - \widehat{RV}\right)^2$
- Quasi-likelihood (QL): $\left[\frac{RV}{\widehat{RV}} - \log\left(\frac{RV}{\widehat{RV}}\right) - 1\right]$
- The standard HAR model and GHAR is often estimated via ordinary least squares (OLS), i.e. EC = MSE.
- When the errors $u_{i,t}$ in HAR are independent, homoscedastic, and normally (Gaussian) distributed, the OLS estimator is consistent under the asymptotic sense.
- Nonetheless, given the stylized facts of RV (such as spikes, heteroskedasticity, and so on), the OLS estimator may not be an ideal choice and a better estimator may be available.

Estimation Criterion (EC)

- ARCH, GARCH, et al: QL
- HAR ([Corsi, 2009]): MSE

- Models trained with QL $\leftrightarrow$ Multiplicative Error Model (MEM) of [Engle, 2002].
- Let y_t be a non-negative random variable, such as RV_t. y_t follows a MEM model if it can be expressed as:

$$y_t = \nu_t z_t \quad \nu_t = g\left(\theta; \mathcal{F}_{t-1}\right) \quad z_t \overset{\text{i.i.d.}}{\sim} D^+\left(1, \sigma^2\right). \tag{11}$$

Estimation Criterion (cont.)

- Supposing z_t is Gamma distributed, the negative log-likelihood, after omitting constants, can be expressed as:

$$L(\theta) = \sum_{t=1}^{T} \left[\frac{y_t}{\nu_t} + \log\left(\nu_t\right) \right]. \tag{12}$$

$$\Rightarrow \left[\frac{RV_t}{\widehat{RV}_t} + \log(\widehat{RV}_t) \right]$$

- Hence, the comparison between models trained with MSE and QL essentially boils down to the comparison between additive models and multiplicative models:

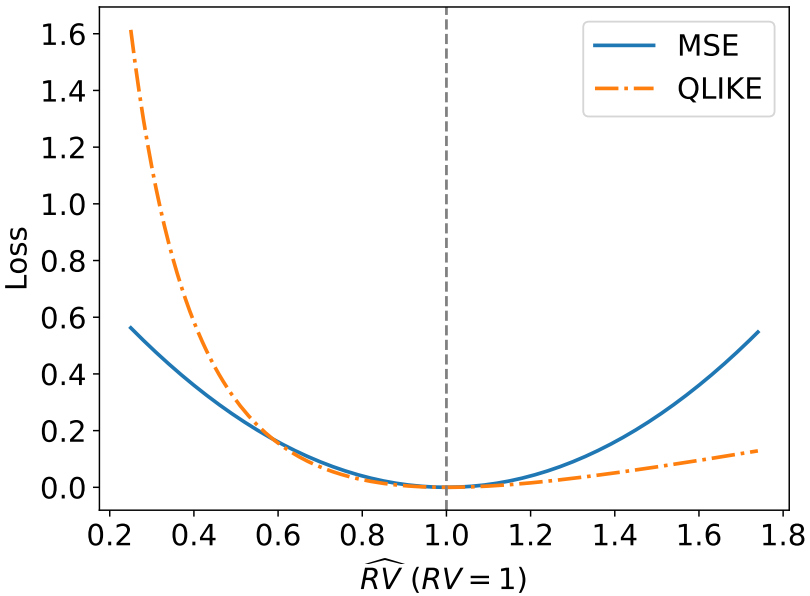
$$RV_t = \begin{cases} \mathbb{E}\left(RV_t \mid \mathcal{F}_{t-1}\right) + u_t, & u_t \text{ zero mean,} \\ \mathbb{E}\left(RV_t \mid \mathcal{F}_{t-1}\right) \times z_t, & z_t \text{ unit mean.} \end{cases}$$

Estimation Criterion (cont.)

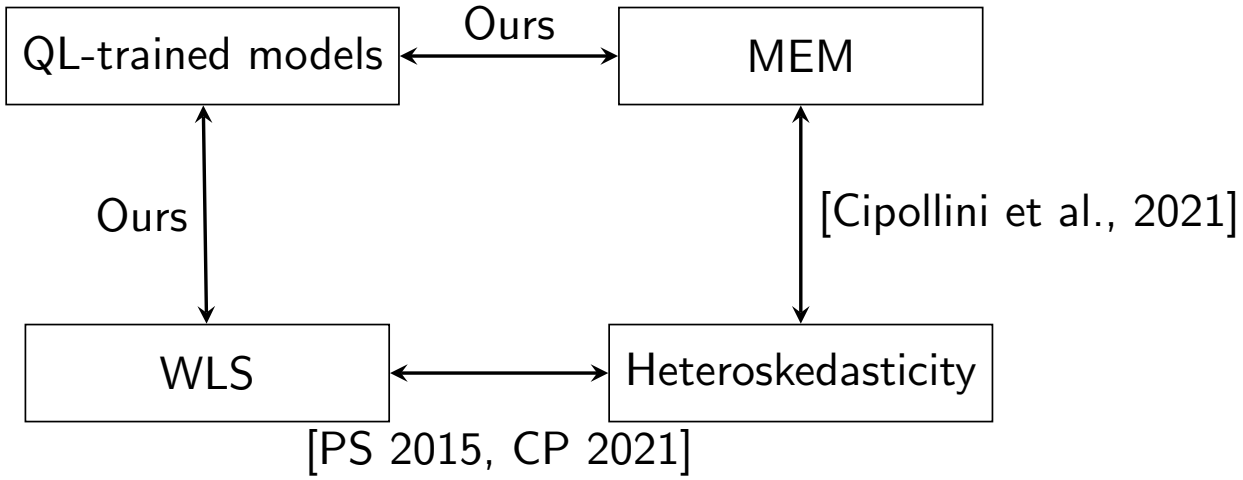
- Additive models $\rightarrow$ heteroskedasticity of errors (u_t)
- Multiplicative models $\rightarrow$ z_t is more homoskedastic ([Cipollini et al., 2021])

- [Patton and Sheppard, 2015] and [Clements and Preve, 2021] proposed to estimate HAR models using different schemes of **weighted least squares (WLS)** to assign less importance during estimation to periods where volatility is less precisely estimated.

Estimation Criterion (cont.)



- Compared to MSE, QL exhibits asymmetry and imposes a higher penalty on under-predictions, which is **useful for RV forecasting during turbulent periods**.
- QL has no analytical solution. Stochastic gradient descent (Adam) is applied.



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Setup

- Universe: DJIA30 and S&P100
- Full Period: July 1, 2007 to Jun 30, 2021.
- All of the models are re-estimated every month based on a rolling sample window of the past 1000 days (around 4 years).
  - 36-month data for model training; the recent 12-month data for validation; the following month for testing.
  - [July 2007, June 2010]; [July 2010, June 2011]; [July 2011].
  - [Aug 2007, July 2010]; [Aug 2010, July 2011]; [Aug 2011].
- Out-of-sample period: 10 years, from July 1, 2011 to June 30, 2021.
- Other choices of setups do not significantly alter our findings.

Forecast Evaluation Approaches

- Forecast loss (FL) is similar yet distinct from EC, as FL evaluates the performance of RV forecasts in out-of-sample testing while EC is used for in-sample training ([Cipollini et al., 2020]).
  1. MSE.
  2. QL.
- (i) Model Confidence Set (MCS) identifies a subset of models with significantly superior performance from model candidates, at a given level of confidence ([Hansen et al., 2011]).
- (ii) Diebold-Mariano (DM) test examines whether there are significant differences between two time-series forecasts ([Diebold and Mariano, 1995]).

Out-Of-Sample Forecast Loss

	1-Day		1-Week		1-Month	
	MSE	QL	MSE	QL	MSE	QL
HAR_M	1.000	1.000	1.000	1.000	1.000	1.000
$GHAR_M$	0.927	0.983	0.904	0.987	0.975*	1.036
$GNNHAR1L_M$	0.907	0.979	0.940	0.943	1.021	0.968
$GNNHAR2L_M$	0.967	0.977	1.034	0.953	1.134	1.032
$GNNHAR3L_M$	1.210	0.982	1.014	0.961	1.046	0.958
HAR_Q	0.927	0.981	0.939	0.945	1.069	0.986
$GHAR_Q$	0.886	0.983	0.842*	0.936	1.151	0.954*
$GNNHAR1L_Q$	0.867*	0.961*	0.855	0.913*	1.179	0.965
$GNNHAR2L_Q$	0.879	0.959*	0.873	0.920	1.736	0.947*
$GNNHAR3L_Q$	0.894	0.963	1.185	0.942	1.502	0.971

Note: The table reports the ratios of forecast losses of various models compared to the standard HAR_M model over the 1-day, 1-week, and 1-month horizons, respectively. For each horizon, the model with the best out-of-sample performance in MSE (QL) is highlighted in red (blue), respectively. An asterisk (*) indicates models that yield as accurate forecasts as the best model at the 5% significance level based on the MCS test.

- GNNHAR1L improves the performance of the linear models.
- The performance of GNNHAR starts to decline when additional GNN layers are added, particularly with three layers.
- QL might serve as a more effective in-sample estimation criterion than MSE.
- Longer-term forecasting becomes less sensitive to graph information.

Calm Periods

	1-Day		1-Week		1-Month	
	MSE	QL	MSE	QL	MSE	QL
	Panel A: Bottom 90%					
HAR_M	1.000	1.000	1.000	1.000	1.000	1.000
$GHAR_M$	0.961	0.998	0.949	1.001	0.967	1.027
$GNNHAR1L_M$	0.943*	0.998	0.883*	0.960*	0.923*	0.924*
$GNNHAR2L_M$	0.944*	0.990	0.901	0.954*	0.946*	0.921*
$GNNHAR3L_M$	0.957	0.987	0.911	0.965	0.937*	0.930*
HAR_Q	1.010	0.984	1.005	0.955*	1.159	0.942*
$GHAR_Q$	0.989	1.007	1.076	1.001	1.257	1.084
$GNNHAR1L_Q$	0.967	0.978*	0.944	0.943*	1.478	0.977
$GNNHAR2L_Q$	0.976	0.979*	0.985	0.947*	1.433	0.973
$GNNHAR3L_Q$	0.970	0.980*	1.062	0.957	1.662	0.969

Note: The bottom 90% (Panel A) and the top 10% (Panel B) RV of the S&P500 ETF index.

- $GNNHAR1L_M$ and $GNNHAR2L_M$ appear to be the most effective based on the MSE loss; while when evaluating accuracy in terms of QL, the models $GNNHAR1L_Q$ and $GNNHAR2L_Q$ provide the most precise forecasts.
- For stationary time series, the alignment of the training loss (i.e. EC) and the testing loss (i.e. FL) may produce improved forecasts.

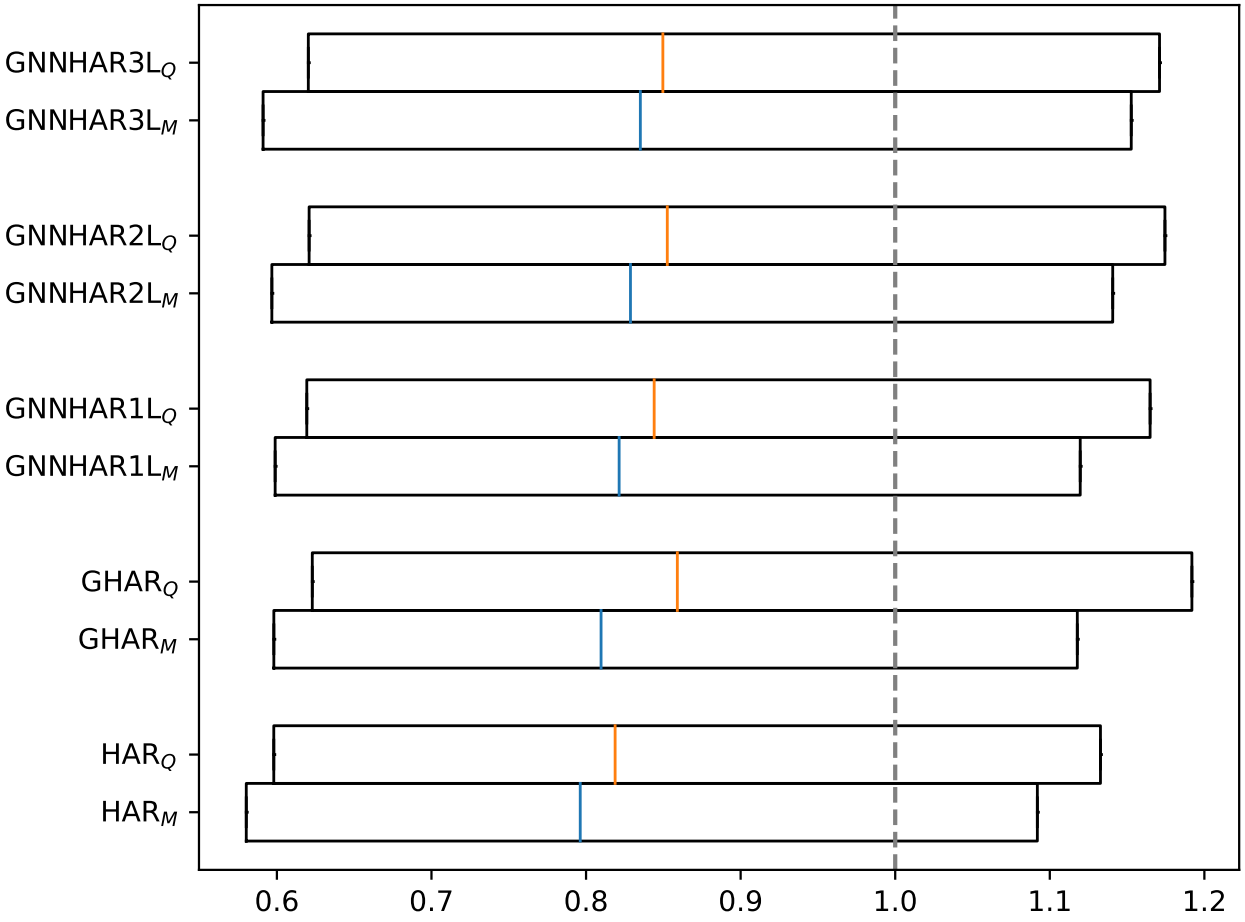
Turbulent Periods

	1-Day		1-Week		1-Month	
	MSE	QL	MSE	QL	MSE	QL
			Panel B: Top 10%			
HAR_M	1.000	1.000	1.000	1.000	1.000	1.000
$GHAR_M$	0.916	0.910	0.897	0.959	0.976*	1.043
$GNNHAR1L_M$	0.895	0.903	0.949	0.908	1.033	1.007
$GNNHAR2L_M$	1.102	0.915	1.056	0.951	1.157	1.131
$GNNHAR3L_M$	1.293	0.958	1.030	0.952	1.059	0.982
HAR_Q	0.900	0.965	0.928	0.925	1.059	1.024
$GHAR_Q$	0.852	0.867*	0.804*	0.799*	1.149	0.841*
$GNNHAR1L_Q$	0.834*	0.879	0.841	0.848	1.143	0.955
$GNNHAR2L_Q$	0.848	0.862*	0.924	0.861	1.773	0.886
$GNNHAR3L_Q$	0.868	0.882	1.205	0.909	1.483	0.973

- We observe the superiority of QL-trained models (especially $GHAR_Q$) over MSE-trained models.
- QL-trained models are less prone to excessive under-prediction during turbulent periods, when compared to the MSE-trained models.

Impact of QL: Turbulent days

Figure 2: Forecast ratios $(\widehat{RV}/RV)$



Note: This figure presents a box plot illustrating three summary statistics: the median, Q1 and Q3 quantiles. Each group consists of two sets of box plots, with the top (resp. bottom) set representing models utilizing QL (resp. MSE) as EC.

Larger Universe: S&P100

	1-Day		1-Week		1-Month	
	MSE	QL	MSE	QL	MSE	QL
HAR_M	1.000	1.000	1.000	1.000	1.000	1.000
$GHAR1L_M$	0.948	0.988	0.909	0.994	0.972*	0.986
$GNNHAR1L_M$	0.963	0.986	0.951	0.944	1.027	1.092
$GNNHAR2L_M$	1.072	0.988	1.031	0.954	1.092	1.000
$GNNHAR3L_M$	1.061	0.986	1.029	0.959	0.992	0.967
$GNNHAR4L_M$	1.047	0.992	1.042	0.975	1.079	0.978
$GNNHAR5L_M$	1.090	0.997	1.057	0.986	1.109	1.038
HAR_Q	0.949	0.983	0.937	0.947	1.171	0.991
$GHAR_Q$	0.919	0.984	0.850*	0.922	1.154	0.939*
$GNNHAR1L_Q$	0.917*	0.969	0.858	0.916	1.231	1.017
$GNNHAR2L_Q$	0.915*	0.969	0.909	0.915*	1.206	0.941*
$GNNHAR3L_Q$	0.938	0.966*	1.178	0.968	1.523	0.946
$GNNHAR4L_Q$	0.985	0.970	1.165	0.972	1.563	0.971
$GNNHAR5L_Q$	0.951	0.968	1.193	0.975	1.741	0.989

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Conclusion

- We propose a novel methodology based on graph neural networks to review volatility spillover effects in equities.
- The information from the multi-hop neighbors in the financial graph does not offer a clear advantage in predicting the volatility of any target stock.
- Nonlinear spillover effects help improve the forecasting accuracy of the realized volatility.
- Compared to the usual MSE loss, using Quasi-likelihood as the training loss function yields more accurate volatility forecasts.
- Future directions:
  1. enrich predictor set [Li and Tang, 2021]
  2. complicated temporal dependency
  3. economic benefits, see [Goyenko et al., 2024].

Thank You!

Any Questions?

https://sites.google.com/view/chaozhang94/

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Non-linearity

- Fraction of Variance of model F which is Unexplained (FVU) by the standard HAR_M model: $\text{FVU}_t = \frac{\sum_{i=1}^{N}\left(\widehat{RV}_{i,t}^{(F)} - \widehat{RV}_{i,t}^{(\text{HAR}_M)}\right)^2}{\sum_{i=1}^{N}\left(\widehat{RV}_{i,t}^{(F)} - \overline{RV}_{t}^{(F)}\right)^2}$
- $\text{FVU}_t = 0$ means that the HAR's RV forecasts explain all of the variation in the predicted RVs provided by the model.

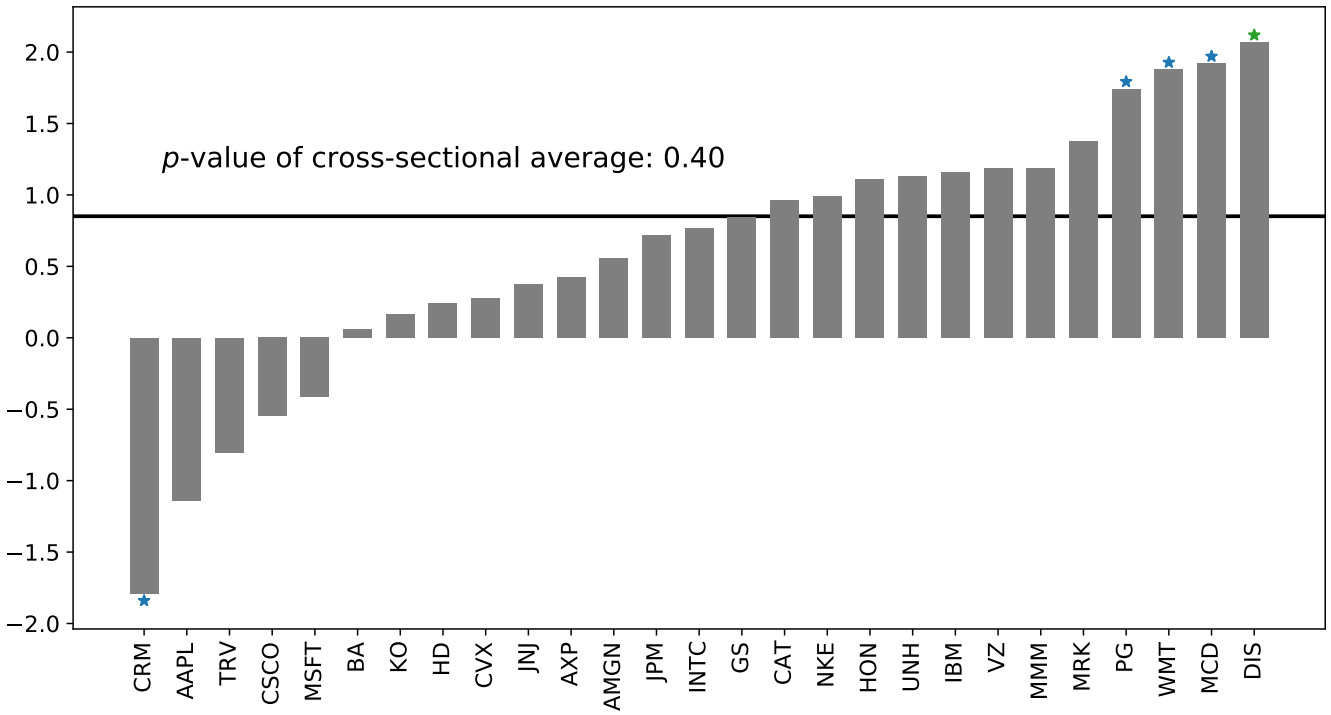
Table 1: FVU compared to HAR_M.

	1-Day		1-Week		1-Month	
	Calm	Turb	Calm	Turb	Calm	Turb
HAR_M	0.000	0.000	0.000	0.000	0.000	0.000
GHAR_M	0.044	0.061	0.054	0.099	0.066	0.092
GNNHAR1L_M	0.077	0.165	0.117	0.244	0.178	0.300
GNNHAR2L_M	0.080	0.205	0.114	0.304	0.207	0.441
GNNHAR3L_M	0.079	0.300	0.130	0.246	0.218	0.272
HAR_Q	0.033	0.056	0.068	0.139	0.184	0.263
GHAR_Q	0.077	0.128	0.108	0.216	0.228	0.779
GNNHAR1L_Q	0.060	0.134	0.102	0.244	0.216	0.886
GNNHAR2L_Q	0.060	0.184	0.118	0.379	0.283	1.391
GNNHAR3L_Q	0.070	0.212	0.163	0.764	0.292	1.236

- Nonlinear models display higher FVU values than linear counterpart GHAR.
- Turbulence $\rightarrow$ More nonlinearity $\rightarrow$ Better performance of GNNHAR

Multi-hop

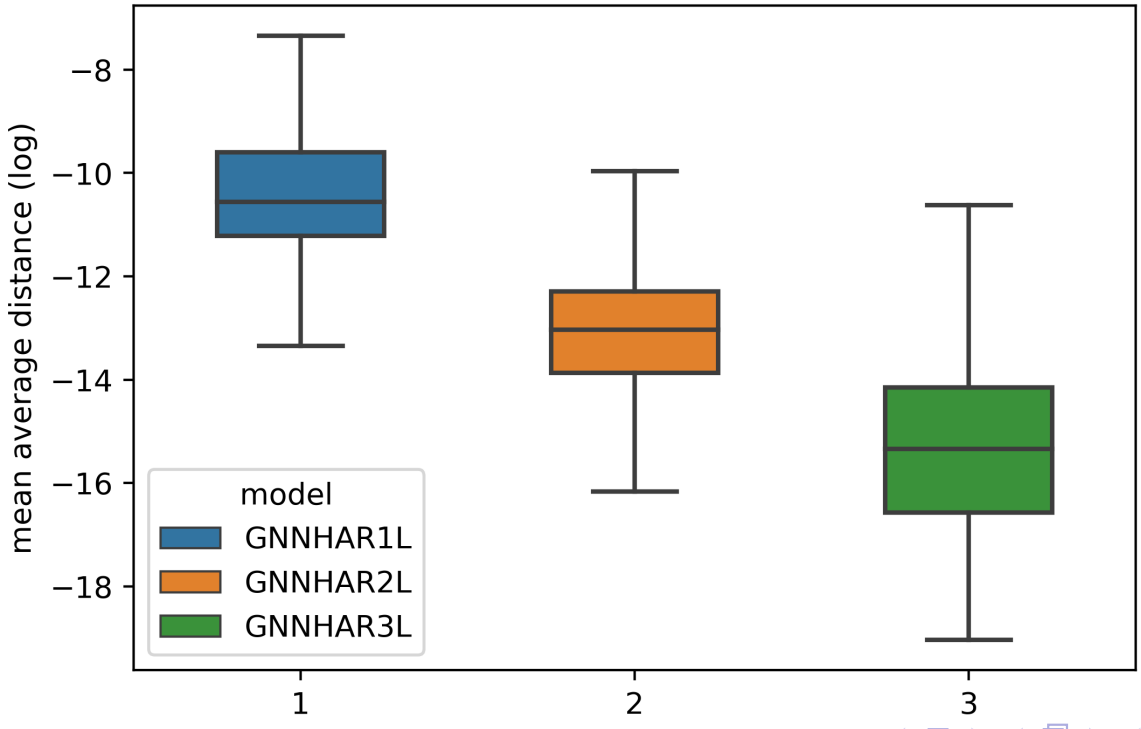
Figure 3: DM test between GNNHAR2L$_Q$ and GNNHAR1L$_Q$.



Note: A positive (negative) number indicates superiority for the GNNHAR1L (GNNHAR2L) model. The y-axis represents the DM test values based on QL between GNNHAR2L and GNNHAR1L, while the x-axis lists the stock symbols. Stars indicate the p-value, with orange, green, and blue representing significance at the 1%, 5%, and 10% levels, respectively. The horizon line represents the cross-sectional DM test value and its corresponding p-value.

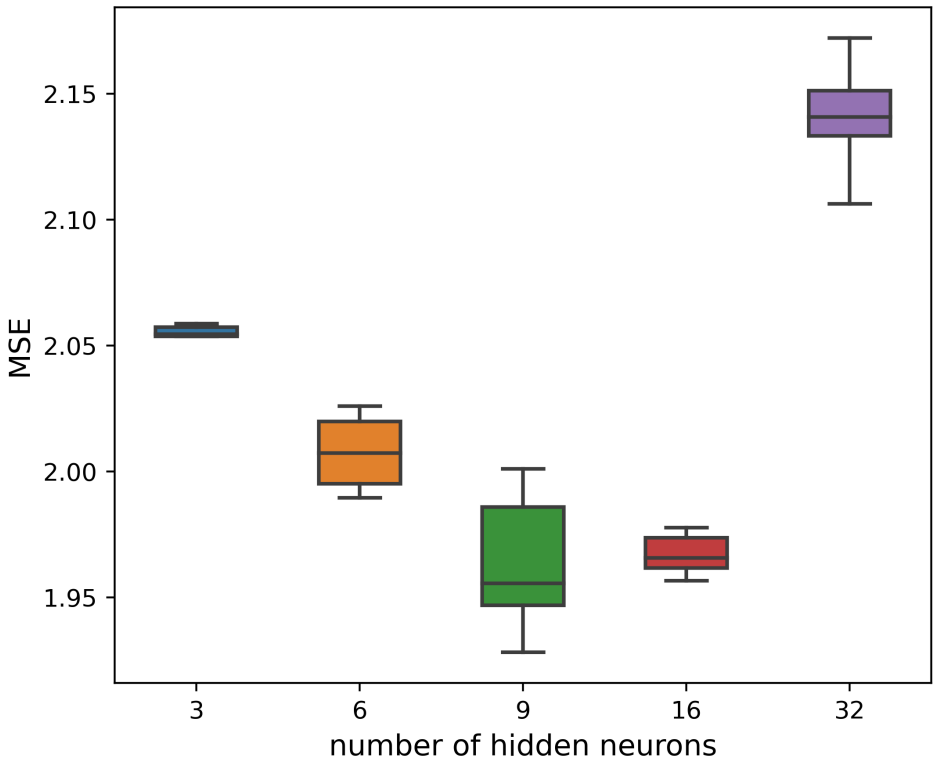
Oversmoothing

- high similarity of node representations obtained at the output layer of GNNs ([Li et al., 2018]).
- stacking with multiple GNN layers that are more than necessary
- Mean Average Distance (MAD [Chen et al., 2020])):
  $\text{MAD} = \frac{\sum_{i=1}^{N} \bar{d}_i}{\sum_{i=1}^{N} \mathbb{1}_{\bar{d}_i > 0}}, \quad \text{where } \bar{d}_i = \frac{\sum_{j=1}^{N} \bar{\boldsymbol{D}}_{ij}}{\sum_{j=1}^{N} \mathbb{1}_{\bar{\boldsymbol{D}}_{ij} > 0}}.$
- $\bar{\boldsymbol{D}}$ is the masked cosine distance matrix, i.e. $\bar{\boldsymbol{D}} = \boldsymbol{D} \circ \boldsymbol{A}$, where $\circ$ denotes the Hadamard product (element-wise multiplication), and $\boldsymbol{D}_{ij} = 1 - \frac{\boldsymbol{H}[i,:] \cdot \boldsymbol{H}[j,:]}{\|\boldsymbol{H}[i,:]\| \|\boldsymbol{H}[j,:]\|}$, where $\boldsymbol{H} = \text{GNN}(\boldsymbol{V}_{:t-1}, \boldsymbol{A})$ is the final layer of GNNHAR.



Hyperparameter Tuning

- Batch size: 32
- Learning rate for Adam: 10^{-3}
- Early stopping
- To determine the optimal hidden dimension, we perform a grid search over $D^{(l)} \in \{3, 6, 9, 16, 32\}$ on validation datasets.



- A hidden dimension of 9 in a one-layer GNNHAR model leads to the smallest MSE on the validation data.
- Similar results are observed for models trained with QL.